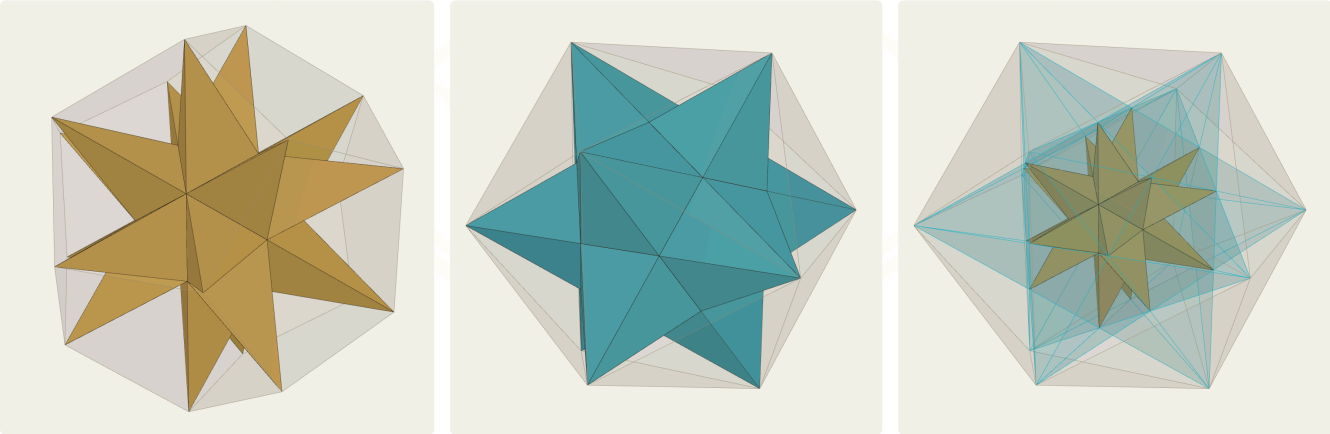
The DICTO Kepler–Poinsot–UP Cycle

The tall companion to Universal PiTet (UP): the great and small stellated dodecahedra in an endless φ^3 cycle, every shell filled exactly by one pattern, in golden reciprocity

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Version 1 · 19 October 2026 · DICTO DOI: 10.5281/zenodo.23464021 · Companion to Universal PiTet (DICTO DOI: 10.5281/zenodo.23463305) · Licence CC BY 4.0 · Scripts and exact checks: https://github.com/DICTOR-Master/kaleidohedra (research branch, studies/2026-10-19-kepler-poinsot-up)

Abstract. Universal PiTet grows from the Cleo star by the tightest solid round the last, and its shells come out thin. DICTO specified a taller version ("think stellation, think Gaudí"): keep every station a true star and fill only between stars. With the icosahedron I of edge 1, the great stellated dodecahedron (spikes to 3× the face distance) has a dodecahedral hull of edge φ, the small stellated dodecahedron on that hull (spires to $\sqrt{5}$× the face distance) has an icosahedral hull of edge φ^3, and the cycle repeats for ever, $\times\varphi^3$ in length and $\times\varphi^9$ in volume each turn. The spires stand φ^2 and φ times their face radius and the hull steps $\times\varphi$ then $\times\varphi^2$, balancing within each turn; the stars are exactly 3× and $\sqrt{5}$× their cores in volume. Every shell is filled exactly by the Cleo star's own pattern: the spikes, inward pyramids from the hull's faces and valley tetrahedra along the core's edges. In the first shell these are DICTO's CLEO, cap and AXE. The two shells meet in golden reciprocity, PiTet's own principle of opposites: the stars' points stand φ^2 and φ times their face radius, and the inward pyramids of the other shell are exactly their reciprocals, $1/\varphi^2$ and $1/\varphi$. All exact in $\mathbb{Q}(\sqrt{5})$.



(a) Kepler's great stellated dodecahedron (gold) in its clear hull, the dodecahedron D1; (b) the small stellated dodecahedron (cyan) on D1, in its clear hull, the icosahedron I·φ^3; (c) both together: one turn of the cycle.

1. The cycle

Let I be the icosahedron of edge 1. On each face raise a pyramid whose tip lies at three times the face's distance from the centre: this is Kepler's great stellated dodecahedron (the spikes' sides fall into its twelve pentagram planes), and its twenty tips are the corners of a dodecahedron D1 of edge φ. On each face of D1 raise a pyramid with its tip at $\sqrt{5}$ times the face distance: Kepler's small stellated dodecahedron, whose twelve tips are the corners of the icosahedron I·φ^3. That icosahedron starts the next turn. The stars and their hulls are classical

[1–3, 13]; their icosahedral order is the same order that frustrates flat-space packings of tetrahedra [20]. What follows is the cycle's proportions and its fill.

	Edge	Spire height ÷ face radius	Star ÷ core (volume)
I (core)	1		
great stellated star on I		φ^2	3 (each spike = I/10)
D1 (its hull)	φ		
small stellated star on D1		φ	$\sqrt{5}$ (12 spires = $(\sqrt{5}-1)\cdot$D1)
I·φ^3 (its hull)	φ^3		one turn: $\times\varphi^9$

Table 1. The proportions of one turn. The step on the triangles (three-fold axes) has the taller spire (φ^2) and the smaller hull step ($\times\varphi$); the step on the pentagons (five-fold) has the shorter spire (φ) and the larger hull step ($\times\varphi^2$). Every turn is the same.

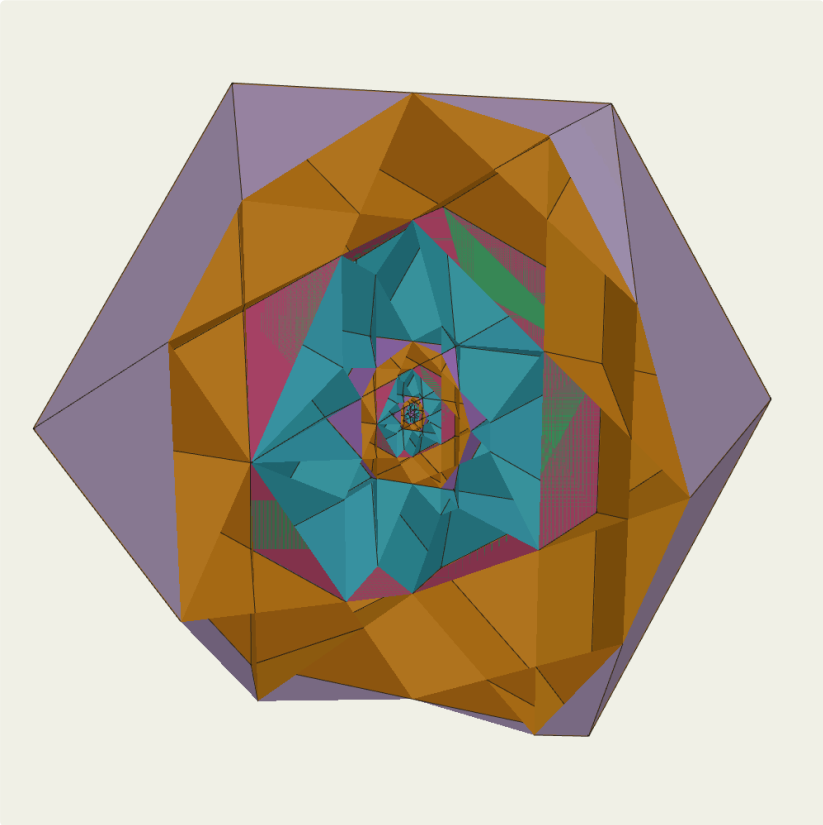


Figure 1. Three turns of the cycle nested, cut away: each turn is $\varphi^3 \approx 4.24$ times the one inside it, the same stars and the same pieces each time; the cycle continues inward and outward without end (a fourth turn inward would be 1/76 of the outer one's size).

2. The fill (exact)

Each shell, core → star → hull, is filled by the pattern of DICTO's Cleo star (DICTO DOI: 10.5281/zenodo.23378070): the star's spikes; one inward pyramid on each face of the hull, its apex at the core's corner beneath; and one valley tetrahedron on each edge of the core, its other two corners the tips of the two spikes beside that edge.

Shell	Pieces	Exact volume of each	Is
I → great star → D1	20 spikes	1/8 + √5/24	DICTO's **CLEO**
	12 inward pyramids	5/8 + 7√5/24	DICTO's **cap**
	30 valley tetrahedra	1/6 + √5/12	DICTO's **AXE**
D1 → small star → I·φ³	12 spires	5/3 + 3√5/4	new here
	20 inward pyramids	29/24 + 13√5/24	new here: **CLEO's reciprocal**
	30 valley tetrahedra	3/4 + √5/3	new here

Table 2. The two shells of a turn. In each, the pieces fill core → hull exactly: the volumes sum to the gap with remainder exactly 0, every piece lies in the hull and outside the core, and no two overlap (an exact separating plane for every close pair). Each kind is one volume; each next turn repeats them φ^3 larger.

The first shell is DICTO's AXE, CLEO and cap kit [9, 10], already published as in-between stages (Capturing the In-betweens, DICTO DOI: 10.5281/zenodo.23345502): icosahedron + 20 CLEO is the great stellated dodecahedron, + 30 AXE the excavated dodecahedron [4], + 12 caps the dodecahedron. CLEO and AXE are Mosseri–Sadoc tiles [5, 6]. The second shell's three pieces are not met before.

Golden reciprocity (exact). Measure each pyramid by its height over the circumradius of its base. The great star's spike (CLEO) stands at φ^2, and the second shell's inward pyramid at exactly $1/\varphi^2$; the small star's spire stands at φ, and the first shell's inward pyramid (the cap) at exactly $1/\varphi$. Each star's points are met, in the other shell, by points turned inward whose height is their golden reciprocal: tall and flat, outward and inward, with products exactly 1 (DICTO's "opposite structures throughout", PiTet's principle).

Point	(height ÷ base radius)²	Its reciprocal, in the other shell	(height ÷ base radius)²
great star's spike (CLEO)	$7/2 + 3\sqrt{5}/2 = \varphi^4$	small shell's inward pyramid	$7/2 - 3\sqrt{5}/2 = \varphi^{-4}$
small star's spire	$3/2 + \sqrt{5}/2 = \varphi^2$	great shell's inward pyramid (cap)	$3/2 - \sqrt{5}/2 = \varphi^{-2}$

Table 3. The reciprocal pairs, exact (exact_reciprocal.py). In each pair the two values are Galois conjugates: $\sqrt{5} \to -\sqrt{5}$ turns one into the other.

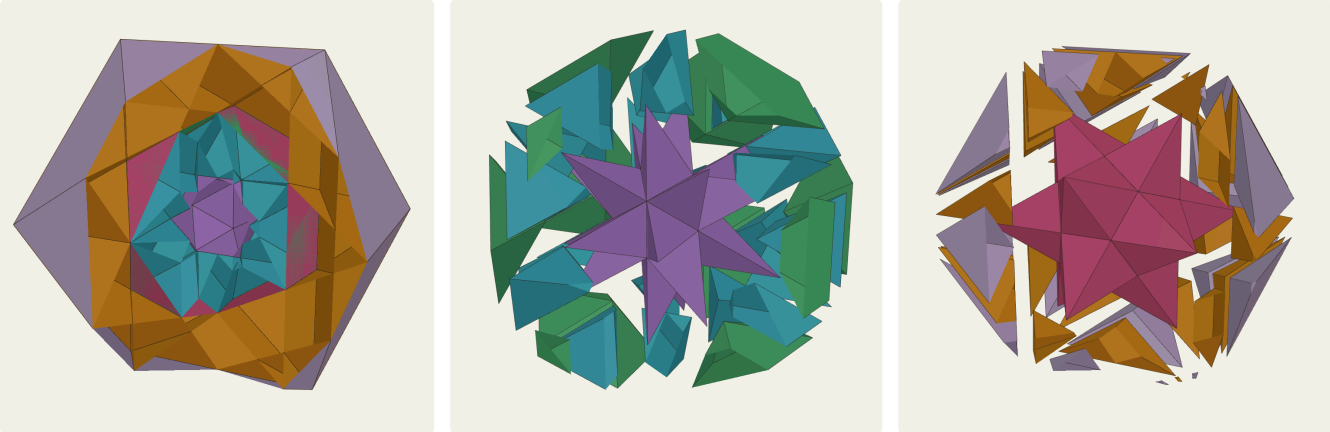


Figure 2. Left: one turn cut away, from the core (gold) out to I·φ³. Middle: the great stellated star (CLEO, lilac) with its caps (green) and AXEs (cyan) pulled away. Right: the small stellated star (spires, rose) on the first shell, with its inward pyramids (pale lilac, CLEO's golden reciprocal) and the new valley tetrahedra (amber) pulled away.

3. A note on perspective

DICTO relates the tall spires to the sculptors' practice of shaping a figure for its viewpoint (Gothic portal figures; Michelangelo's David, made for a place high above the viewer). A pinhole-camera model (float) gives two results. Seen from outside, on a spire's base plane at six base radii, the true spires already look almost right (corrections ×0.94 and ×1.03). Seen from inside, standing in a valley at a spike's foot, they look two to three times squatter (2.9× for the great star, 2.1× for the small one), and no height reads right, because added height recedes from view. In the Kaleidohedra app a reverse-perspective projection (REV) lets a viewer inside a shell see it open out instead.

4. Prior art and what is claimed

Classical. The great and small stellated dodecahedra, their hulls and the φ^3 step belong to Kepler [1] and Poinsot [2] and are part of the regular polyhedra [3, 13]; the great stellated dodecahedron's volume, three icosahedra, follows from the standard formulas. Stellation theory in general is set out by Messer for the rhombic triacontahedron [14]. Kepler already set the regular solids between concentric spheres [15].

Mosseri–Sadoc tiles and inflation. CLEO and AXE are Mosseri–Sadoc tiles [5]. Mosseri and Sadoc gave inflation rules for their tiles with factor τ; Papadopolos and Ogievetsky decorated them and derived the inflation matrix [16], and showed that the six golden tetrahedra admit no stone inflation for any power of τ [17]. Kramer reviews the central icosahedral fillings these come from [6]. Closest to this paper, Koca et al. tile icosahedral patches face to face with composite Mosseri–Sadoc tiles under an inflation whose volume eigenvalue is τ^3, and find dodecahedra of edge 1 and τ in the second and third inflations [18]; Al Raisi et al. tile dodecahedra inflated by τ^n with modified Mosseri–Sadoc tiles [19]. Those are tilings that inflate; the cycle here is different in kind: Kepler's two stars nested without end, each shell filled once by the Cleo star's pattern, the stars kept whole.

Not found in the sources checked (the full text of the 87 library sources in the appendix, and web searches on 19 October 2026): the endless star cycle filled by one pattern; the small stellated dodecahedron as $\sqrt{5}$ times its core; the second shell's fill, and the golden reciprocity between the stars' points and the inward pyramids. Related dissections of the stellated dodecahedra exist [7].

Outlook: the reciprocal cycle. Reciprocation about the centre (corners ↔ face planes) takes each star to its classical dual, the great stellated dodecahedron to the great icosahedron and the small stellated dodecahedron to the great dodecahedron, and each hull to its dual. Because the cycle repeats by similarity, its reciprocal is a cycle of the other two Kepler–Poinsot solids, also gaining φ^3 a turn. It brings no new solids; whether its shells fill by one pattern is not yet checked.

Data and reproducibility

exact_kp.py (exact: the cycle, the volumes, the fill and no-overlap); kp.py, cleo_pattern.py (float exploration), perspective.py (the perspective model); export_figs.py and the figure page.

References

1. J. Kepler, Harmonices Mundi (1619).

2. L. Poinsot, Mémoire sur les polygones et les polyèdres, J. École Polytechnique (1810).
3. H. S. M. Coxeter, Regular Polytopes, 3rd ed., Dover (1973).
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5. R. Mosseri, J. F. Sadoc, Two and three dimensional non-periodic networks obtained from self-similar tiling, in: The Structure of Non-Crystalline Materials (1982).
6. P. Kramer, Gateways towards quasicrystals, arXiv:1101.0061 (2011).
7. I. Hafner, dissections of the small and great stellated dodecahedra into a rhombic triacontahedron and a rhombic hexecontahedron, Visual Mathematics 9(2) (2007); Dissection of Two Stellated Dodecahedra, Wolfram Demonstrations.
8. DICTO, Universal PiTet: The DICTO PiTet Proposition, DICTO DOI: 10.5281/zenodo.23463305.
9. DICTO, AXE, CLEO and FUJI: the Gap Pieces of the Icosahedron, DICTO DOI: 10.5281/zenodo.23351291.
10. DICTO, Spheres of Dharma, DICTO DOI: 10.5281/zenodo.23310833.
11. DICTO, Theaetetus, DICTO DOI: 10.5281/zenodo.23378070.
12. DICTO, Capturing the In-betweens, DICTO DOI: 10.5281/zenodo.23345502.
13. M. Cutler, E. Schulte, Regular polyhedra of index two, I (2011).
14. P. W. Messer, Stellations of the rhombic triacontahedron and beyond, Structural Topology 21 (1995).
15. J. Kepler, Mysterium cosmographicum (1596).
16. Z. Papadopolos, O. Ogievetsky, On inflation rules for Mosseri–Sadoc tilings (1999).
17. O. Ogievetsky, Z. Papadopolos, On quasiperiodic space tilings, inflation and Dehn invariants (1999).
18. N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi, Dodecahedral structures with Mosseri–Sadoc tiles (2020).
19. R. Al Raisi, N. O. Koca, M. Koca, R. Koc, Modified Mosseri–Sadoc tiles from D6 (2026).
20. J.-F. Sadoc, R. Mosseri, Geometrical Frustration, Cambridge University Press (1999).

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md). The full text of the 87 sources below was searched on 19 October 2026 for this paper's subjects: the Kepler–Poinsot solids and stellation; Mosseri–Sadoc tiles, golden tetrahedra and their inflations; nested and concentric polyhedra; dodecahedra and icosahedra built from tiles.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov	1885	Elements of the Theory of Figures (Nachala ucheniya o figurah)
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
R. Buckminster Fuller	1954	Building construction (geodesic dome)
R. Buckminster Fuller	1961	Synergetic building construction (octet truss)
R. Buckminster Fuller	1962	Tensile-integrity structures
Kenneth D. Snelson	1965	Continuous tension, discontinuous compression structures
Peter J. Pearce	1971	Synthesized natural geometric structures
Stephen C. Baer	1973	Structural system
John P. Hogan	1976	Homohedral construction employing icosahedron
Peter J. Pearce	1976	Minimum inventory maximum diversity building system
Haresh Lalvani	1988	Building structures based on polygonal members and icosahedral symmetry
Helmut Bergman	1988	Continuous spherical truss construction (US 4,719,726)
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
H. Lalvani	1995	Periodic space structures composed of two nodal polyhedra for design applications (US 5,448,868)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
H. Lalvani	1997	Building systems with non-regular polyhedra based on subdivisions of zonohedra (US 5,623,790)
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra

M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
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